

2024, Mar. 2024 Release LOC:Acute Pediatric

Diabetes Mellitus

Utilization Benchmarks

Length of Stay: 1.9 d
Length of Stay Type: InterQual
Condition/Procedure: DIABETES MELLITUS
% Paid as Observation: 68%

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview

Select Day

- Episode Day 1 (1, 2)
- Episode Day 2 (1, 2)
- Episode Day 3 (2, 32)
- Episode Day 4

Discharge Screens (34)

Notes

InterQual® criteria (IQ) is confidential and proprietary information and is being provided to you solely as it pertains to the information requested. IQ may contain advanced clinical knowledge which we recommend you discuss with your physician upon disclosure to you. Use permitted by and subject to license with Change Healthcare LLC and/or one of its subsidiaries. IQ reflects clinical interpretations and analyses and cannot alone either (a) resolve medical ambiguities of particular situations; or (b) provide the sole basis for definitive decisions. IQ is intended solely for use as screening guidelines with respect to medical appropriateness of healthcare services. All ultimate care decisions are strictly and solely the obligation and responsibility of your health care provider. © 2023 Change Healthcare LLC and/or one of its subsidiaries. All Rights Reserved.

Overview

Instruction: The Diabetes Mellitus (DM) subset is for patients with new-onset DM and pre-existing DM who are experiencing hyperglycemia.

- For patients with diabetic ketoacidosis, refer to the Diabetic Ketoacidosis subset
- For patients with hypoglycemia, refer to the Hypoglycemia subset
- For patients with pregestational DM or gestational DM, refer to the Antepartum subset

Diabetes mellitus (DM) is a complex chronic condition that requires ongoing medical care, multiple interventions, and risk reduction strategies to improve outcomes. Type 1 diabetes, type 2 diabetes, diabetes due to other causes, and gestational diabetes mellitus (GDM) are the four general classification categories of DM. Type 1 DM typically leads to complete deficiency of insulin, is caused by autoimmune beta cell destruction, and can occur in adulthood as latent autoimmune diabetes. Loss of sufficient beta cell insulin secretion causes type 2 DM, typically due to insulin resistance. Chemical-induced and drug-induced DM, exocrine pancreas diseases, and monogenic diabetes syndromes represent types of diabetes that originate from other causes. The clinical presentation and disease progression of type 1 and type 2 DM can significantly differ. Both type 1 DM and type 2 DM can occur in children and adults. To determine the treatment regimen, the classification of DM is essential. However, at the time of diagnosis, some patients cannot be distinctly classified into one of these categories and misdiagnoses can occur. As time progresses, beta cell deficiency in patients becomes more evident. Multiple genetic and environmental factors cause loss of beta cells or beta cell function, resulting in hyperglycemia. The risk of acute complications such as diabetic emergencies and long-term microvascular and macrovascular complications increases in all people with diabetes once hyperglycemia develops (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

Evaluation and Treatment:

Diagnostic tests for diabetes mellitus (DM) include an A1C value greater than or equal to 6.5% (48 mmol/mol), fasting

plasma glucose value greater than or equal to 126mg/dL (7.0mmol/L) measured when there has been no caloric intake for a minimum of 8 hours, or a two-hour plasma glucose value greater than or equal to 200 mg/dL (11.1 mmol/L) obtained during a 75g oral glucose tolerance test. A diagnosis of diabetes may occur when a patient with symptoms of hyperglycemia or hyperglycemic crisis has a random plasma glucose value greater than or equal to 200 mg/dL (11.1 mmol/L). In patients with no symptoms of hyperglycemia, the diagnosis requires two abnormal screening test results from the same or separate samples. Knowing the plasma glucose level and confirming that symptoms are a result of diabetes is vital to making informed decisions related to treatment. To improve inpatient management and discharge planning, hospitalized patients with DM should have A1C measured if not performed within the previous three months. When possible, consulting and collaborating with providers and teams trained in diabetes management is encouraged to reduce hospital length of stay, reduce readmission following hospitalization, improve outcomes, and enhance glycemic control. Parent/caregiver and patient involvement are vital for successfully implementing the prescribed treatment regimen and self-management plan for pediatric patients with diabetes. The pediatric diabetes team should work with the patient and parents/caregivers to evaluate all factors and potential barriers impacting the treatment plan. The patient and parent/caregiver should also receive individualized and developmentally appropriate diabetes self-management education, support, and assistance to overcome barriers and establish new goals as appropriate. Clinical judgment and ongoing assessment and evaluation of clinical statuses, such as the severity of illness, comorbidities, nutritional intake, medications that may affect glucose levels, and changes and trends in measured glucose levels, are necessary to make individualized treatment decisions (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Glaser et al., *Pediatr Diabetes* 2022, 23: 835-56).

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:** ^(1, 2)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽³⁾● Diabetes mellitus, **Both:**● **Finding, One:**

- Inability to tolerate or adhere to prescribed regimen (including diet) ^(4, 5, 6)
- New onset ⁽⁷⁾

● **Intervention, All:** ^(8, 9)

- Blood glucose monitoring ^(10, 11)
- Diabetes education ^(12, 13)
- Insulin or oral glucose-lowering medication initiation or adjustment ^(14, 15, 16)

● **Hyperglycemia and, Both:**● **Finding, ≥ One:**● **Creatinine, ≥ One:** ⁽¹⁷⁾

- ≥ 1.5x baseline and chronic kidney disease ⁽¹⁸⁾
- ≥ 1.5x upper limit of normal ^(19, 20)
- Increased by ≥ 0.3 mg/dL(26.5 μmol/L) within last 2d
- Intractable nausea and vomiting ⁽²¹⁾

● **Orthostatic hypotension, Both:** ⁽²²⁾

- Sustained drop in blood pressure within 3 min of sitting or standing

● **Blood pressure, ≥ One:**

- Systolic drop ≥ 20 mmHg
- Diastolic drop ≥ 10 mmHg

● **Intervention, All:** ^(8, 9)

- Blood glucose monitoring ^(10, 11)
- Diabetes education ^(12, 13)
- Insulin or oral glucose-lowering medication initiation or adjustment ^(14, 15, 16)
- IV fluid ^(23, 24, 25)

● **Episode Day 2, One:** ^(1, 2)● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁶⁾

- Blood glucose within acceptable limits ^(27, 28)
- Patient or caregiver knowledgeable about disease and understands the regimen
- Tolerating PO ⁽²⁹⁾

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**● **Blood glucose, All:**● **Finding, One:**● **Glucose, ≥ One:**

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L) ⁽³⁰⁾
- Blood glucose monitoring ^(10, 11)
- Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ^(14, 15, 16)
- Inadequate oral intake and IV fluid ^(23, 24, 31)
- Patient or caregiver knowledge deficit and diabetes education ^(12, 13)

● **Episode Day 3, One:** ^(2, 32)● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁶⁾

- Blood glucose within acceptable limits ^(27, 28)
- Patient or caregiver knowledgeable about disease and understands the regimen
- Tolerating PO ⁽²⁹⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care

for this condition, **One:**

- For diabetes mellitus use Episode Day 3 at a higher level of care
- For a condition other than diabetes mellitus use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

● **ACUTE, One:** ⁽³³⁾

● **Blood glucose, All:**

● **Finding, One:**

● **Glucose, ≥ One:**

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L) ⁽³⁰⁾

- Blood glucose monitoring ^(10, 11)
- Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ^(14, 15, 16)
- Patient or caregiver knowledge deficit and diabetes education ^(12, 13)

● **Episode Day 4, One:**

● **ACUTE, One:**

● **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽²⁶⁾

- Blood glucose within acceptable limits ^(27, 28)
- Patient or caregiver knowledgeable about disease and understands the regimen
- Tolerating PO ⁽²⁹⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, One: ⁽³⁴⁾

● **HOME, Both:**

● **Level of care appropriateness, Both:**

- Home environment safe and accessible ⁽³⁵⁾
- Patient or caregiver demonstrates ability to manage care

● **Diabetes mellitus discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁶⁾

● **Education on the importance of, All:**

- Prevention and treatment of hypoglycemia
- Home monitoring of blood glucose
- Sick day management
- Foot inspection and care
- Annual influenza vaccine
- Annual eye examination
- Lifestyle modification to prevent hypertension and dyslipidemia ⁽³⁷⁾
- Smoking cessation, if indicated

⁽³⁸⁾ Glycemic control and hypertension management to prevent retinopathy, nephropathy, and neuropathy

- Medication reconciliation ⁽³⁹⁾
- Follow-up care planned ⁽⁴⁰⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● **Level of care appropriateness, All:**

- Home environment safe and accessible ⁽³⁵⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(41, 42)

● **Skilled services, ≥ One:** ⁽⁴³⁾

- Skilled nursing ⁽⁴⁴⁾

- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴⁵⁾
- **Diabetes mellitus discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽³⁶⁾
 - **Education on the importance of, All:** ⁽⁴⁶⁾
 - Prevention and treatment of hypoglycemia
 - Home monitoring of blood glucose
 - Sick day management
 - Foot inspection and care
 - Annual influenza vaccine
 - Annual eye examination
 - Lifestyle modification to prevent hypertension and dyslipidemia ⁽³⁷⁾
 - Smoking cessation, if indicated
 - Glycemic control and hypertension management to prevent retinopathy, nephropathy, and neuropathy ⁽³⁸⁾
 - Medication reconciliation ⁽³⁹⁾
 - Follow-up care planned ⁽⁴⁰⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - **Level of care appropriateness, One:**
 - Support systems available ⁽⁴⁷⁾
 - **Skilled treatment, ≥ One:**
 - Clinical assessment ⁽⁴⁸⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁴⁹⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴⁴⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(50, 51)
 - Functional impairments requiring at least supervision ⁽⁵²⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴⁴⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(50, 51)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵²⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(50, 51)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵²⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽³⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(50, 51)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵²⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁵³⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽³⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● Level of care appropriateness, **Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵⁴⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(55, 56)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵⁷⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽³⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Notes:**1:****Expected progress after 24h**

- Blood glucose level improved
- Vital signs stable and within acceptable limits
- PO intake improving
- Activity increasing

Consider:

- Changing medications to PO or stabilizing insulin regimen for discharge
- Nutrition consultation
- Discharge planning
- Continued treatment of precipitating event

2:**Care Facilitation**

Potential risk for readmission

- Knowledge deficit or non-adherence with condition management identified (**Home care, Readmission Reduction Program**)

Social Determinants of Health (SDOH)

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Children and caregivers should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the child and family unit (Kreuter et al., Annu Rev Public Health 2021, 42: 329-44; Sokol et al., Pediatrics 2019, 144:). The following social needs should be assessed on admission to determine which interventions are relevant to the individual child and their family support system. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given child and family should be planned for throughout the hospital stay.

*Interventions should be considered for all patients and families if social need is relevant

Alcohol or substance use

- Coordinate care with outpatient primary care or behavioral health providers*
- Refer to addiction social support systems*
- Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication-assisted treatment)
- Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting women with substance use disorders aftercare) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
- Provide information about Cognitive Behavioral Therapy (CBT), Multidimensional Family Therapy (MDFT), and/or Pharmacotherapy for youth with co-occurring emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

Child maltreatment, abuse, or intimate partner abuse - sexual and/or psychological abuse

- Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe*
- Assist with creating a long-term safety plan or make appropriate referral*
- Refer to child protective services (CPS) and other domestic violence services if warranted*
- Refer for psychosocial support if warranted*
- Provide community resources and legal information
- Provide information on parent training programs such as the Nurse-Family Partnership Program (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

End of life planning, arrange palliative care or hospice

- Consider palliative care for children with end-stage disease
- Consider hospice care for terminally ill children with a life expectancy of less than 6 months

Environmental exposure to toxic substances and other physical hazards

Lead poisoning:

- Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Tobacco use (e.g., cigarettes, E-cigarettes, vaping):

- Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021)

Financial resources strained or limited

{tab}Food insecurity:

- Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Berman et al., Pediatr Rev 2018, 39: 235-46)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, fuel, or rent supplements)

{tab}Income insecurity:

- Identify income resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Infants and children with Special Health Care Needs

- Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)*
- Refer for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues, (e.g., bronchopulmonary dysplasia or feeding dysfunction (Pediatrics 2008, 122: 1119-26)*)
- Refer for adaptive/assistive medical equipment and equipment for technology-dependent children, (e.g., home oxygen, ventilators, monitors, or tube feeding) (Pediatrics 2008, 122: 1119-26)*
- Refer for neurodevelopmental assessment for high-risk infants*
- Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start Program for infants or toddlers under the age of 3 and pregnant women in need of medical, mental health, nutrition, or education assistance, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the child's second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
- Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
- Refer for respite services, if warranted

Mental health co-morbidities involved

- Arrange appointment with primary care provider or behavioral health specialist upon discharge*

- Coordinate care with outpatient behavioral health providers*
- Provide information about suicide prevention, including the National Suicide Prevention hotline*
- Refer for psychiatric services for suicide risk screening and intervention
- Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
- Provide information about Intensive Community-based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management)
- Transfer to day treatment program, residential addiction services or partial hospitalization when warranted

Risk for treatment nonadherence

- Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers, such as text message reminders to take medication and electronic medication dispensers
- Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- Refer to interventions to prevent medical relapse or readmission, and to enhance medication compliance

Risk for readmission

- Consider transfer to an alternative level of care such as subacute care
- Consider home care services
- Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) (J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
- Consider intensive in-home behavioral health services
- Provide patient and family with condition-specific information including appropriate planning for prevention of RSV infection and appropriate immunizations

Special Health Risk Behaviors in adolescence and young adulthood

- Arrange follow-up appointment with primary care provider and/or specialists upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information on mentoring programs, (e.g., Big Brothers Big Sisters of America)
- Refer for intensive community-based treatment for children with comorbid behavioral health issues
- Refer to community resources for teen pregnancy prevention/family planning (use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

For eating disorders (e.g., bulimia):

- Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18;; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)
- Provide information on outpatient psychosocial interventions and family-based treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

For parenting resources:

- Provide information on Family Resource Centers and Parenting support groups for young, first-time parents, (e.g., National Parent Helpline) (National Parent and Youth Helpline)
- Provide information on community resources to improve parenting skills (face-to-face or by telephone) (U.S. Department of Health and Human Services, Parenting Resources. 2021)

Unstable living environment or homelessness expected upon discharge

- Assist with health insurance application
- Provide information about homeless shelter application to caregivers if applicable
- Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to federally sponsored outpatient programs
- Refer to intensive case management intervention, assertive community treatment, or other intensive community treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Application of this criteria point is appropriate for patients who cannot tolerate or adhere to their prescribed treatment regimen, including those who fail outpatient treatment.

5:

Social determinants of health can cause significant health inequities related to diabetes and an inability to adhere to the treatment regimen which may contribute to complications and poor outcomes. Financial barriers such as medications and device costs contribute to medication nonadherence and poorly controlled glycemic levels. Food insecurity and the inability to consistently obtain nutritious foods are associated with a higher risk for uncontrolled glycemic levels, decreased medication adherence, sub-optimal self-care behaviors, and depression. Housing insecurity can often coincide with food insecurity, decreased health literacy and health numeracy deficiencies, lack of insurance coverage, mental illness, and cognitive dysfunction. Patients with diabetes and home insecurity require secure places to keep and store diabetes medications and supplies to adhere to their prescribed regimen. Language barriers can contribute to an increased risk of diabetes complications in non-English speakers. Incorporating language assistance and offering education materials in multiple languages can reduce language barriers and improve these patients' communication, knowledge, adherence, and health outcomes. Communication of diabetes care and education should be presented in a manner that is easy to comprehend. Health numeracy requires numeric skills such as interpreting food labels and blood glucose levels for medication administration. Health literacy and numeracy correlate with patients' ability to manage their disease and self-care effectively. They also contribute to a patient's understanding of diabetes management and complication prevention concepts and are essential to establishing and promoting adherence to a treatment regimen (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

6:

Malfunction and failure of diabetes technology can represent barriers that contribute to an inability to tolerate and adhere to the diabetes medical management plan; early recognition and management are essential. Diabetes technology refers to devices, software, and hardware used to enhance diabetes condition management, quality of life, and the health of people with diabetes. The complexity and changing landscape of diabetes technologies requires planning, training, education, and follow-up to prevent complications from these barriers. Additional education needs and potential barriers should be assessed periodically in case of complications and unmet goals and/or outcomes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

7:

Individuals with newly diagnosed diabetes may be unaware of the clinical symptoms of diabetes mellitus, such as polyuria, nocturia, polydipsia, blurred vision, and polyphagia, which can result when significant hyperglycemia

surpasses the renal glucose threshold. Increased plasma glucose levels may cause glycosuria, increased urine osmolality, and subsequent diuresis. Increased diuresis contributes to polyuria and nocturia. Fluid losses stimulate thirst and lead to polydipsia and consequent fluid shifts involving the eye's lens can cause blurred vision. Also, the substantial loss of calories in the urine may cause polyphagia. Lastly, insulin deficiency accelerates protein catabolism and the breakdown of fat, all of which can lead to unintentional weight loss (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Winter et al., In: Winter et al., ed. Handbook of Diagnostic Endocrinology, 3rd edn. 2021:238-9).

8:

A primary consideration for admitting or discharging a patient with diabetes (i.e., new-onset diabetes mellitus (DM) or pre-existing DM) is ensuring that the patient has the knowledge, tools, and caregiver support (if applicable) to effectively manage this condition at home, even in the absence of complicating factors. Adults who are socioeconomically disadvantaged, older, and/or uninsured, and patients with mental health conditions have significant risks and barriers that complicate outpatient management. Assessing for potential hazards and obstacles (e.g., inability to obtain medications and supplies, knowledge deficits related to blood glucose monitoring and insulin administration) is vital (Maloney and Glauser, In: Walls, ed. Rosen's Emergency Medicine: Concepts and Clinical Practice, 10th edn. 2022:1550-1). Safe and quality DM management and discharge planning involve assessing and identifying risks and barriers at admission and addressing them before discharge.

9:

The goals of diabetes treatment are to prevent and delay complications and enhance the patient's quality of life. Patient-centered glycemic management includes educating patients and their families or caregivers to promote informed and shared decision-making. A patient-centered focus considers the patient's preferences, chooses a regimen that optimizes tolerance and adherence, empowers the patient, and ensures diabetes self-management education and support access. Glycemic targets are a necessity to include in the patient-centered strategy. Simplifying the treatment plan can improve self-efficacy and the patient's ability to meet target goals. If needed, target goal adjustments are appropriate to improve safety and adherence to the treatment regimen (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Davies et al., Diabetes Care 2018, 41: 2669-701).

10:

The target glycemic range for non-critically ill hospitalized patients with diabetes is 100-180 mg/dL (5.6-10.0 mmol/L). Blood glucose concentrations greater than 180 mg/dL (10.0 mmol/L) may be acceptable, at times, to reduce the risk of harmful outcomes in patients with severe comorbidities or decreased life expectancy. Blood glucose targets should be individualized and modified if there are identifiable risks for adverse effects such as a history of frequent hypoglycemia or hypoglycemia unawareness (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

11:

In non-critically ill hospitalized patients with adequate nutritional intake, recommendations are to complete point-of-care (POC) glucose testing immediately before meals. Alternatively, in patients with inadequate oral intake, oral intake restrictions, or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing every 4-6 hours. However, blood glucose monitoring is individualized. Additional or more frequent blood glucose testing is necessary if the patient cannot effectively communicate their needs, recognize signs or symptoms of a complication, or has a history of increased risk of diabetes-related adverse events (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

12:

National standards recommend diabetes self-management education and support (DSMES) for patients and/or caregivers to enhance knowledge, decision making, and mastery of skills for diabetes management to prevent acute complications, improve treatment outcomes, and reduce risk of long-term complications. Hospitalization represents a critical time to evaluate, assess, and provide DSMES that includes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Identifying a health care provider to provide diabetes care post discharge
- Establishing home blood glucose goals, self-monitoring of blood glucose, and understanding when to call the provider
- Recognition, treatment, and prevention of hyperglycemia and hypoglycemia

- Administration of blood glucose-lowering medications, including insulin, if relevant
- Information on healthy dietary choices
- Proper use and disposal of insulin needles and syringes
- Sick-day management (e.g., when sick with a cold, flu, or infection)

13:

Critical events such as a new diagnosis, unmet treatment goals, complications, and transitions in life require evaluation of diabetes self-management educational needs to encourage acquiring the skills that may assist in executing and adhering to the treatment regimen and medical nutrition therapy (MNT) plan. An individualized MNT plan created through a joint effort of shared decision making with the patient and the healthcare team that includes a registered dietitian nutritionist is essential. The goals of MNT are to attain individualized glycemic targets, delay or prevent complications, reach and maintain body weight goals, incorporate personal and cultural preferences, address barriers to behavioral changes and access to healthy foods, and provide the tools and resources to develop and maintain healthy dietary patterns. Individualized macronutrient recommendations consider the patient's eating patterns, preferences, and metabolic goals, as evidence shows there are no general recommendations for macronutrients for all people with diabetes. Carbohydrate monitoring strategies are essential to achieve glycemic control and include, but are not limited to, carbohydrate counting, an exchange system, or experience-based estimation (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American Diabetes Association, Diabetes Care 2022, 45:S1-S262).

14:

Instruction: Medication adjustment includes initiation of a new drug, change in dosage, and/or administration frequency of a medication. Blood glucose values guide medication dosing decisions and adjustments. Administration of sliding scale insulin must occur at least 3 times within 24 hours. A continuous infusion rate must have at least 2 rate adjustments within 24 hours (including insulin pump rate adjustments). A glucose-lowering medication dose withheld based on normal or low blood glucose levels is included in the count of doses administered within the 24-hour timeframe.

15:

Scheduled subcutaneous insulin is recommended to manage hyperglycemia in non-critically ill hospitalized patients. The preferred treatment for patients with adequate nutritional intake includes an insulin regimen that consists of basal, prandial, and correction insulin aligned with meals. Recommendations are to complete point-of-care (POC) glucose testing and subcutaneous rapid or short-acting insulin administration immediately before meals. Alternatively, in patients with inadequate oral intake, prandial insulin administration is recommended directly after the patient eats, with adjustments in dosing related to the amount of food consumed. The preferred treatment for patients with oral intake restrictions includes basal insulin or basal plus correction insulin. For these patients or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing and subcutaneous rapid or short-acting insulin administration every 4-6 hours (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

16:

When making treatment decisions, providers should assess for socioeconomic and psychosocial barriers and consider medication-taking behavior and the religious and cultural preferences of the patient. For non-critically ill patients with hyperglycemia and type 2 diabetes mellitus (DM) in the hospital, insulin is often the preferred treatment. However, based on patient preference, it can be appropriate to give oral glucose-lowering medication. If deemed appropriate by the provider, patients with preexisting DM may continue their home regimen, including oral glucose-lowering medication. When oral glucose-lowering medication is on hold during hospitalization, resumption of this medication may be suggested 1-2 days prior to discharge (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

17:

Risk for acute kidney injury increases in individuals with diabetes. Impaired kidney function increases the potential for difficult glycemic control and treatment-related hypoglycemia challenges. The presence of chronic kidney disease (CKD) impacts the risks and benefits associated with rigorous glycemic control and many glucose-lowering medications. Adverse effects such as hypoglycemia and mortality are more likely to occur in patients with CKD (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Kidney Disease: Improving Global Outcomes

Diabetes Work, Kidney Int 2022, 102: S1-S127).

18:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

19:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

20:

Creatinine elevation reflects renal function impairment. The following values are equivalent to 1.5x upper limit of normal (ULN) by age: (McMillan et al., The Harriet Lane Handbook. 2008)

Age/Creatinine

- 3 days to 1 year: ≥ 0.6 mg/dL (53 μ mol/L)
- > 1 year to 3 years: ≥ 0.9 mg/dL (80 μ mol/L)
- > 3 years to 11 years: ≥ 1.1 mg/dL (97 μ mol/L)
- 12 years to < 18 years: ≥ 1.5 mg/dL (133 μ mol/L)

21:

Vomiting and pronounced hyperglycemia necessitate adjustment of the treatment regimen and expeditious interaction with a diabetes care team. A patient managed with noninsulin pharmacological agents or medical nutrition only may require insulin in addition to ensuring sufficient fluid and caloric intake (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

22:

Orthostatic hypotension (OH) is a sustained decrease in systolic blood pressure ≥ 20 mmHg or a decrease in diastolic blood pressure ≥ 10 mmHg within 3 minutes of standing or a head-up tilt (HUT) test (Fedorowski et al., Circ Arrhythm Electrophysiol 2022, 15: e010573). The orthostatic stress tests of active standing or HUT assess for postural changes in blood pressure. However, the HUT test has limitations because it requires a tilt table which may not be routinely available in all clinical settings. The sit-up test is used to overcome the limits of the HUT test, requires no equipment, and assesses postural changes in those who may have impaired balance or challenges with upright standing (Oyake et al., Blood Press Monit 2022, 27: 79-86). In the patient with diabetes, OH can be correlated with diabetic cardiac autonomic neuropathy, diabetes duration, uncontrolled diabetes, volume depletion, dehydration, and certain medications (Idiaquez Rios et al., J Diabetes Complications 2022, 36: 108169).

23:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are $\geq 50\%$ (0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):

Weight ≤ 10 kg = ≥ 50 mL/kg/24h

- Hourly rate = 2.08 (mL/h) x kg

Weight > 10-25 kg = ≥ 40 mL/kg/24h

- Hourly rate = 1.67 (mL/h) x kg

Weight > 25-60 kg = ≥ 30 mL/kg/24h

- Hourly rate = 1.25 (mL/h) x kg

Example: An 11 kg child is required to receive a minimum of 40 mL/kg/24h to meet this criteria point. Multiply 1.67 (mL/h) x 11 (child's weight in kg) = 18.37 mL/h. This is the minimum fluid replacement rate for this child.

24:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

25:

Patients with uncontrolled diabetes and symptomatic hyperglycemia are at risk for acute complications that necessitate fluid repletion; complications include, but are not limited to, dehydration. Dehydration is more likely to require hospitalization in individuals with diabetes than in those without diabetes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

26:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

27:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

28:

A glucose target between 100-180 mg/dL (5.6-10.0 mmol/L) is recommended for most patients in noncritical care units (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

29:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg/d*) can be calculated based on current body weight (*kg*):

- 100 *mL/kg/d* for first 10 *kg*
- 50 *mL/kg/d* for second 10 *kg*
- 20 *mL/kg/d* for each additional *kg* in weight

30:

The risk of hypoglycemia increases in young children (less than six years of age) who often cannot recognize or communicate their needs or manage hypoglycemia. There is a strong correlation between the frequency of blood glucose monitoring and glucose management. Therefore, confirmation of hypoglycemia may require frequent blood glucose monitoring in young children and infants (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Abraham et al., *Pediatr Diabetes* 2018, 19 Suppl 27: 178-92).

31:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

32:**Expected progress after 48-72h**

- Blood glucose stable or close to baseline
- Adequate PO intake
- Activities at baseline
- Preparing for discharge within the next 24 hours
- Patient or care giver demonstrates ability to manage condition

If no, consider:

- Endocrinology consultation for diabetic adjustment for uncontrolled diabetes mellitus or gastroparesis management
- Diabetic educator consultation
- Infectious disease consultation for persistent fever/infection
- Continued treatment of precipitating event

33:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

34:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

35:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

36:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

37:

Cardiovascular disease (CVD) is the major cause of mortality and morbidity in patients with diabetes mellitus (DM). A blood pressure level less than 130/80 mmHg in adolescents aged 13 years or older or less than the 90th percentile for age, sex, and height is recommended in individuals with diabetes to reduce CVD mortality. Patients with DM should be educated about lifestyle modifications that effectively reduce hypertension and dyslipidemia. These modifications may include (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Weight loss, if overweight
- Dietary Approaches to Stop Hypertension (DASH), including reducing sodium and increasing potassium intake
- Reduction of saturated fat, trans fat, and cholesterol consumption
- Increase of n-3 fatty acids, viscous fiber, and plant stanols/sterols consumption
- Moderation of alcohol consumption
- Increase in physical activity

Initial therapy for abnormal lipids should focus on controlling blood glucose levels and medical nutrition therapy that consists of limiting calories from fat to 25-30%(0.25-0.30) and saturated fat less than 7%(0.07), a cholesterol daily limit less than 200mg/day, obtaining 10%(0.10) of calories from monounsaturated fats, and avoiding trans fats. Statin therapy may be considered for youth after 10 years of age who continue to have LDL cholesterol greater than 160 mg/dL (4.1 mmol/L) or LDL cholesterol greater than 130 mg/dL (3.4 mmol/L) and one or more CVD risk factors despite lifestyle and nutrition modifications. Therapy goals aim to achieve an LDL cholesterol value less than 100 mg/dL (2.6 mmol/L) (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

38:

Large prospective randomized studies have shown that intensive diabetes management to achieve near-normal glucose levels delays the onset of nephropathy and diabetic retinopathy. Improved glycemic control has also been shown to modestly slow the progression of neuropathy. Reducing hypertension is also an effective way of preventing the progression or development of nephropathy and retinopathy (American Diabetes Association,

Diabetes Care 2024, 47: S1-S314).

39:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

40:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

41:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

42:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

43:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

44:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

45:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

46:

Reference: (American Diabetes Association, Diabetes Care 2024, 47: S1-S314)

47:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

48:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

49:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

50:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

51:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

52:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when

patient's safety awareness is impaired

- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

53:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

54:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

55:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

56:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

57:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility

